

Hari Ram Selvaraj

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Education

Syracuse University , School of Information Studies, MS in Applied Data Science (<i>GPA 3.93</i>)	May 2026
● Relevant Coursework: Machine Learning, Deep Learning, NLP, Statistical Reasoning for Data Science, SQL, Human-Centered AI	
Bangalore North University , New Horizon College, Bachelor of Computer Applications	Oct 2022

Experience

Data Scientist Intern (GenAI / LLM Systems) - ProHealth Home Health & Hospice	May 2025 - Present
● Engineered an Ask-Me-Anything (AMA) retrieval-augmented SQL intelligence system over Snowflake and Postgres, reducing manual effort by 45% while delivering execution-grounded answers with near-zero SQL hallucinations.	
● Designed semantic schema retrieval using YAML metadata with Arctic embeddings, retrieving top-50 columns at cosine ≥ 0.7 to constrain generation and reduce schema errors by $>90\%$.	
● Deployed multi-model routing across AWS Bedrock, SageMaker, and Vertex AI via LiteLLM, cutting cost per run 35%-48% while maintaining p95 latency 3.8-4.6s.	
● Built a clinician-facing Clinical Overview RAG pipeline with deterministic chunking and hybrid retrieval, achieving 98.8% claim faithfulness and 99.2% citation completeness.	
● Increased self-serve adoption by 30% and accelerated reporting turnaround by 25% through guided prompts and workflow productization for end users.	
Data Scientist - Tata Elxsi	Jan 2023 - Aug 2024
● Designed forecasting and anomaly detection for large-scale streams, improving demand reliability by 35%.	
● Built scalable data pipelines processing 4M records to support AI-driven decision workflows, reducing latency by 40%.	
● Implemented automated experimentation and validation frameworks measuring system impact, increasing engagement metrics by 18%.	
Data Scientist Intern - Tata Elxsi	Aug 2022 - Dec 2022
● Conducted EDA on operational datasets to identify predictive features and modeling opportunities, improving feature readiness for downstream modeling.	
● Built baseline regression and classification models with cross-validation and error analysis, improving benchmark performance by 10%-15%.	
● Produced dashboard-ready datasets and visual summaries for product and operations teams, enabling data-backed prioritization decisions.	
Research Data Scientist - NEXIS Technology Lab	Sep 2024 - Present
● Built a HIPAA and PHI safe, citation-grounded healthcare LLM using Snowflake and Postgres plus hybrid RAG, achieving 92% faithfulness and 4.7/5 clinician usefulness.	
● Productionized governance and deployment with PHI and injection defenses, sustaining approximately 0% PHI leakage.	
● Developed automated evaluation and red-team test suites for PHI leakage and citation failures, improving release confidence and preventing regressions in production.	

Projects

Enterprise Support Knowledge RAG + Agent Assist	2025
● Built an enterprise-grade hybrid RAG system over 1,200+ support documents (3M+ tokens) using section-aware chunking, BM25 + vector search, and cross-encoder reranking, improving Recall@10 from 0.62 to 0.84.	
● Reduced hallucination rate from 8% to $<2\%$ by enforcing citation-bound generation with confidence gating and cite-or-abstain logic.	
● Optimized p95 latency from 4.8s to 2.1s and lowered cost per query from \$0.11 to \$0.04 through context compression, reranker top-3 selection, and embedding cache reuse.	
● Ran a 3-week A/B pilot with 12 support agents, reducing average handle time by 22% and lowering escalation rates by 14%, generating approximately 25 labor-hours per day in operational savings.	
Bitcoin Flash Crash Prediction	2025
● Built a flash crash classifier using Reddit sentiment plus OHLCV features with imbalance handling and calibration, improving minority-event recall from 0.42 to 0.71 at a tuned threshold.	
● Aligned sentiment signals to minute-level timestamps and reduced subreddit dimensionality using PCA, improving validation AUC by 0.08 and stabilizing variance across market regimes.	
● Improved model generalization using time-based splits, regularization, and threshold tuning with calibration, reducing overfitting and improving stability across unseen market regimes.	

Skills

Languages: Python, SQL (Advanced), R

GenAI / LLM Systems: RAG, embeddings, semantic search, vector databases (Pinecone, OpenSearch), LoRA, QLoRA

ML: XGBoost, Random Forest, Logistic Regression, SHAP, time-series modeling, PyTorch, TensorFlow, Scikit-learn, LSTM

Data: Snowflake (Cortex), PostgreSQL, BigQuery, ETL, Apache Spark

Cloud / MLOps: AWS Bedrock, SageMaker, GCP Vertex AI, Azure Cognitive Search, Docker, FastAPI

Conversational AI: Rasa, Dialogflow

Visualization: Tableau, Matplotlib